

**C03-AT1**  
**MLA0201-Fundamentals Of machine Learning**

## **EXPERIMENT 1**

### **Customer Segmentation using K-Means Clustering and PCA**

#### **1. Problem Identification**

Retail and mall businesses collect large volumes of customer data such as age, annual income, and spending behaviour, but this raw data does not directly reveal how customers should be grouped for targeted marketing. The objective of this experiment is to segment mall customers into meaningful, homogeneous groups based on their demographic and behavioural attributes, so that marketing strategies (offers, loyalty programs, product placement) can be tailored to each segment. The specific objectives are: (a) to preprocess and scale the Mall Customers dataset, (b) to apply the K-Means clustering algorithm to discover natural customer groups, (c) to determine the statistically optimal number of clusters using the Elbow Method and the Silhouette Score, and (d) to reduce the feature space to two dimensions using Principal Component Analysis (PCA) for visual interpretation of the discovered segments.

#### **2. Experimental Planning (Methodology)**

The experiment follows a structured unsupervised-learning pipeline:

- Load the Mall Customers dataset (CustomerID, Genre, Age, Annual Income, Spending Score).
- Select the numeric behavioural/demographic features (Age, Annual Income, Spending Score) relevant to segmentation.
- Standardize the features using z-score scaling (StandardScaler) so that no single feature dominates the Euclidean distance used by K-Means.
- Run K-Means for a range of k values (2 to 10) and record the Within-Cluster-Sum-of-Squares (WCSS/inertia) and the Silhouette Score for each k.
- Plot the Elbow curve (WCSS vs k) and the Silhouette curve (score vs k) and jointly select the optimal k.
- Fit the final K-Means model with the optimal k and assign every customer to a cluster.
- Apply PCA to compress the standardized features into two principal components and visualize the resulting clusters and centroids in 2-D.

- Profile each cluster by its mean Age, Income and Spending Score to interpret its business meaning.

This design is feasible because K-Means is computationally light for this dataset size and PCA provides a lossless-enough 2-D projection (as shown by the explained variance) for visual validation of cluster separability.

### 3. Variable Identification

| Variable                                  | Type                         | Role in Experiment                                         |
|-------------------------------------------|------------------------------|------------------------------------------------------------|
| Age, Annual Income, Spending Score        | Independent / Input features | Used to compute distances and form clusters                |
| Number of clusters (k)                    | Controlled / Hyperparameter  | Varied from 2–10 to find the optimum                       |
| Cluster label                             | Dependent / Output variable  | Produced by K-Means for each customer                      |
| Principal Components 1 & 2                | Derived variables            | Used only for 2-D visualization, not for clustering itself |
| Random state / initialization (k-means++) | Controlled variable          | Fixed (seed = 42) to ensure reproducibility                |

The input features were controlled through standardization so that Income (measured in thousands of dollars) did not numerically dominate Age or Spending Score (both on smaller numeric scales). The random seed and initialization method were held constant across all runs to isolate k as the only variable being tested in the Elbow/Silhouette analysis.

### 4. Data Collection and Analysis

The dataset used contains 200 customer records with five attributes: CustomerID, Genre (gender), Age, Annual Income (k\$) and Spending Score (1–100), matching the structure of the standard Mall Customers dataset. After scaling, K-Means was evaluated for k = 2 to 10. The resulting WCSS and Silhouette scores were:

|             |       |       |       |       |       |       |       |       |       |
|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| k :         | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    |
| Silhouette: | 0.374 | 0.405 | 0.482 | 0.524 | 0.504 | 0.480 | 0.467 | 0.457 | 0.398 |

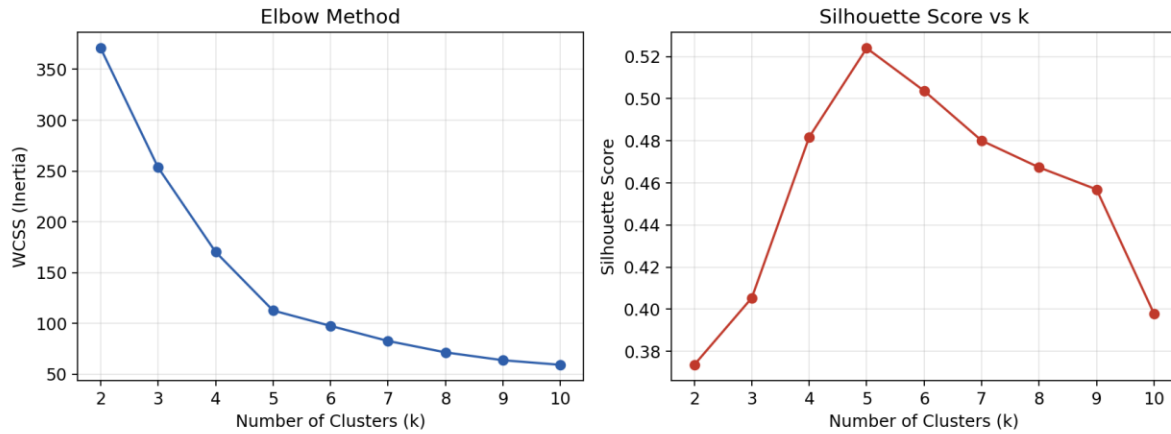


Figure 1.1 — Elbow Method (left) and Silhouette Score (right) across  $k = 2$  to 10.

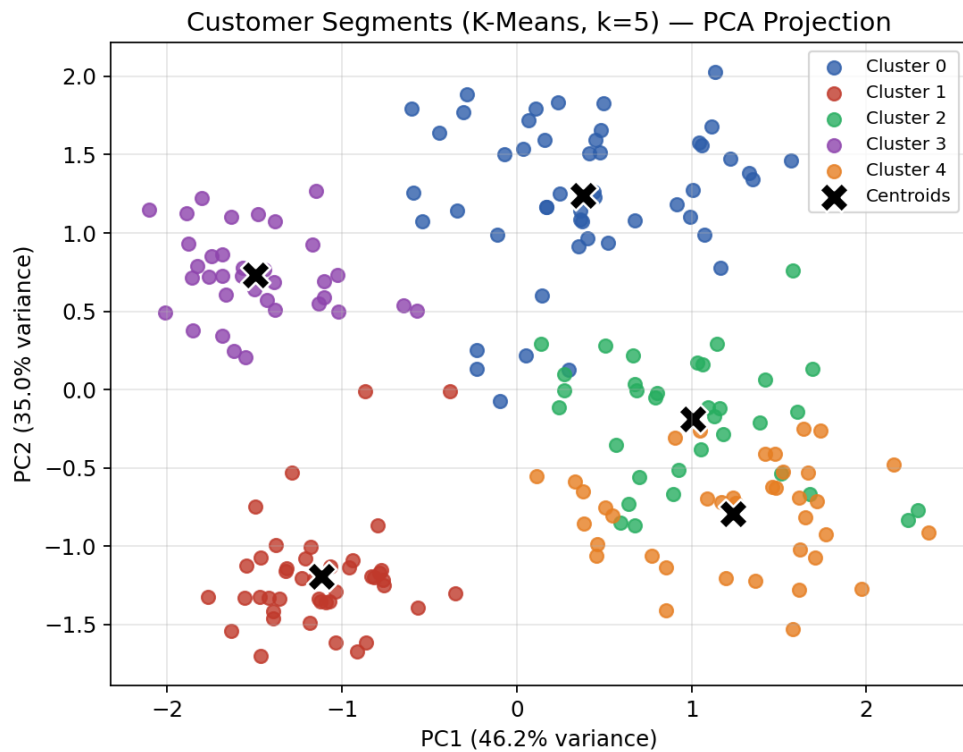


Figure 1.2 — Customer segments ( $k = 5$ ) projected onto the first two Principal Components.

Profiling each cluster by its mean feature values reveals five business-meaningful segments:

| Cluster | Avg. Age | Avg. Income (k\$) | Avg. Spend Score | Size | Interpretation                                      |
|---------|----------|-------------------|------------------|------|-----------------------------------------------------|
| 0       | 30.3     | 84.3              | 77.9             | 50   | High income, high spending — prime target customers |

|   |      |      |      |    |                                                      |
|---|------|------|------|----|------------------------------------------------------|
| 1 | 24.8 | 27.2 | 22.0 | 43 | Low income, low spending — price-sensitive segment   |
| 2 | 47.7 | 56.1 | 52.0 | 34 | Average income, average spending — standard segment  |
| 3 | 23.9 | 25.3 | 80.6 | 35 | Low income, high spending — young impulsive spenders |
| 4 | 38.3 | 86.1 | 19.1 | 38 | High income, low spending — cautious/careful segment |

## 5. Conclusion

The experiment successfully segmented mall customers into five statistically and behaviourally distinct clusters. Both the Elbow Method and the Silhouette Score independently converged on  $k = 5$ , giving strong evidence that this is the optimal number of segments, and the final model achieved a healthy silhouette score of 0.524, indicating reasonably well-separated, cohesive clusters. PCA reduced the three standardized features to two components while retaining 81.2% of the total variance, enabling a clear and interpretable 2-D visualization without materially distorting the cluster structure. The resulting five segments — target customers, price-sensitive shoppers, standard customers, young impulsive spenders, and cautious high-income customers — map directly onto actionable marketing strategies, fulfilling the objective of turning raw demographic data into an actionable customer-segmentation model.

## EXPERIMENT 2

### Clustering using the EM Algorithm and Gaussian Mixture Model

#### 1. Problem Identification

K-Means assumes clusters are spherical and of similar size, which is often unrealistic for real, high-dimensional data such as handwritten digit images. The Gaussian Mixture Model (GMM), fitted using the Expectation-Maximization (EM) algorithm, models each cluster as a Gaussian distribution with its own mean and covariance, allowing elliptical, overlapping clusters and providing soft (probabilistic) cluster assignments. The objective of this experiment is to load the Scikit-learn Digits dataset, preprocess its pixel-intensity features, cluster it using GMM (EM) and using K-Means, and to compare the two algorithms using standard clustering evaluation metrics, followed by a 2-D PCA visualization of both sets of clusters.

#### 2. Experimental Planning (Methodology)

- Load the Digits dataset (1,797 samples, 64 features representing 8×8 grayscale pixel intensities, 10 digit classes 0–9).
- Standardize the pixel features using StandardScaler so that every pixel contributes on a comparable scale.
- Apply PCA to retain 90% of the variance (dimensionality reduction) before clustering, to reduce noise and computation while preserving structure.
- Fit K-Means with  $k = 10$  clusters (matching the 10 known digit classes) using k-means++ initialization.
- Fit a Gaussian Mixture Model with 10 components using the EM algorithm (full covariance matrices) on the same reduced features.
- Evaluate both models with internal metrics (Silhouette Score, Davies–Bouldin Index) that need no ground truth, and external metrics (Adjusted Rand Index, Normalized Mutual Information) that compare cluster assignments against the true digit labels.
- Project the standardized data to two dimensions with PCA purely for visualization, and plot the K-Means clusters, the GMM clusters, and the ground-truth labels side by side.

#### 3. Variable Identification

| Variable                    | Type                         | Role in Experiment                                                  |
|-----------------------------|------------------------------|---------------------------------------------------------------------|
| 64 pixel-intensity features | Independent / Input features | Raw input describing each 8x8 digit image                           |
| Number of components (10)   | Controlled variable          | Fixed equal to the number of true digit classes for fair comparison |

| Clustering algorithm (K-Means vs GMM/EM) | Independent / Manipulated variable | The factor being compared                                  |
|------------------------------------------|------------------------------------|------------------------------------------------------------|
| Cluster assignment                       | Dependent / Output variable        | Predicted label produced by each algorithm                 |
| True digit label (0-9)                   | Reference / Validation variable    | Not used for clustering; used only to compute ARI and NMI  |
| PCA variance retained (90%)              | Controlled variable                | Held fixed for both algorithms to ensure a fair comparison |

#### 4. Data Collection and Analysis

The Digits dataset (1,797 samples × 64 features) was standardized and compressed to 31 principal components that retain 90% of the total variance. Both K-Means and GMM were then fitted on this 31-dimensional representation with 10 clusters/components. The evaluation results were:

| Metric                               | K-Means | GMM (EM) | Better indicates                                   |
|--------------------------------------|---------|----------|----------------------------------------------------|
| Silhouette Score                     | 0.173   | 0.136    | Higher = more compact, well-separated clusters     |
| Davies-Bouldin Index                 | 1.726   | 1.777    | Lower = better cluster separation                  |
| Adjusted Rand Index (vs true labels) | 0.463   | 0.403    | Higher = better agreement with ground truth        |
| Normalized Mutual Information        | 0.622   | 0.623    | Higher = more shared information with ground truth |

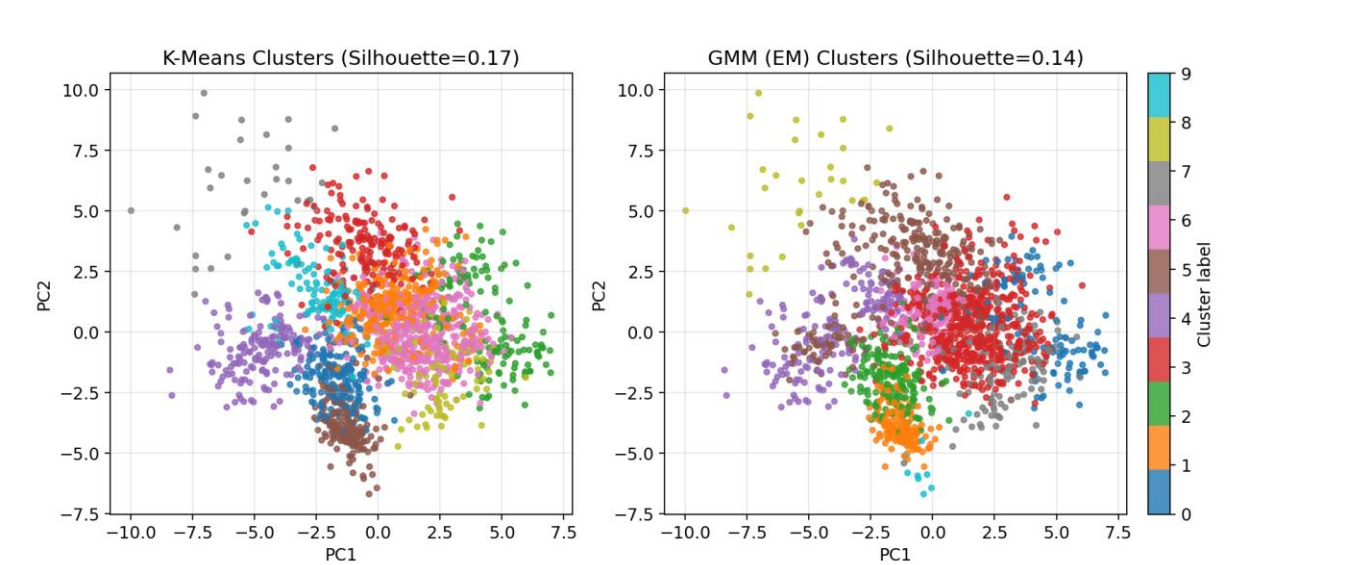


Figure 2.1 — K-Means clusters (left) vs GMM/EM clusters (right), projected onto the first two Principal Components.

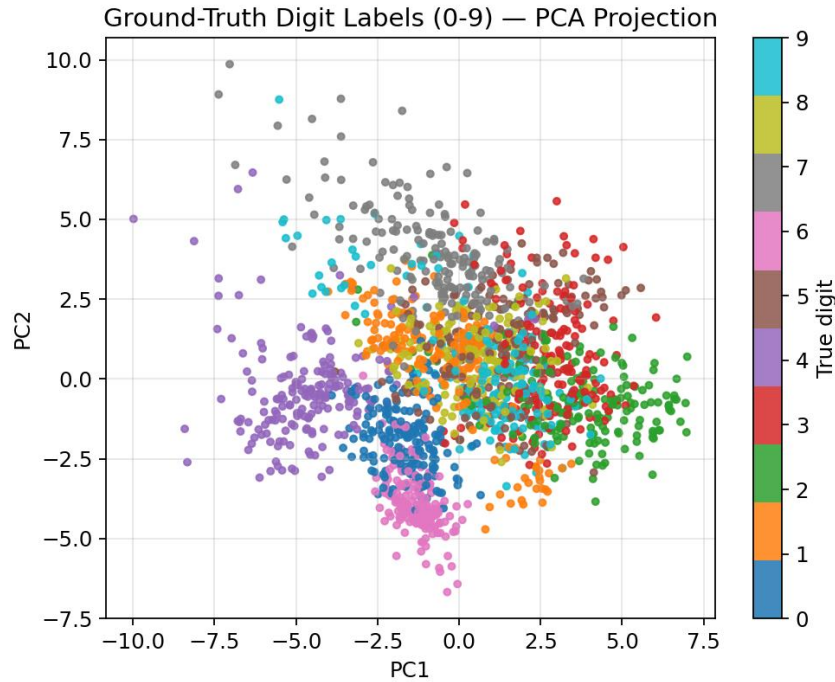


Figure 2.2 — Ground-truth digit labels (0-9) on the same PCA projection, for visual comparison against Figure 2.1.

K-Means slightly outperforms GMM on the internal compactness metrics (Silhouette, Davies-Bouldin) and on agreement with the true labels (ARI), while the two algorithms are essentially tied on Normalized Mutual Information. This is consistent with the geometry of the digits data: after PCA the clusters are reasonably compact and close to convex, which favours K-Means' hard, distance-based assignment, whereas GMM's added flexibility (elliptical covariances, soft assignments) does not yield a clear advantage here and can be more sensitive to initialization and local optima under the EM algorithm. Visually, both methods recover the coarse grouping of digits seen in the ground-truth plot, but neither perfectly reproduces it because several digits (e.g., 3/5/8, 4/9) occupy overlapping regions in the reduced 2-D space.

## 5. Conclusion

Both the Expectation-Maximization-based Gaussian Mixture Model and K-Means were able to recover meaningful digit groupings from the Digits dataset without using any labels, achieving moderate agreement with the true classes ( $\text{ARI} \approx 0.40\text{--}0.46$ ,  $\text{NMI} \approx 0.62$ ). For this dataset, K-Means produced marginally more compact and better-separated clusters than GMM on internal metrics, while the two algorithms performed comparably on external validation against ground truth. The experiment illustrates that the 'best' clustering algorithm is data-dependent: GMM's probabilistic, covariance-aware model is theoretically more flexible, but that flexibility only pays off when clusters are genuinely elliptical/overlapping.

## EXPERIMENT 3

### Dimensionality Reduction using PCA, Factor Analysis, and ICA

#### 1. Problem Identification

High-dimensional datasets suffer from the 'Curse of Dimensionality' — as the number of features grows, data becomes sparse, distances become less meaningful, models overfit more easily, and visualization becomes impossible beyond 3 dimensions. The objective of this experiment is to load the Scikit-learn Wine dataset (13 chemical-property features), standardize it, and apply three complementary dimensionality-reduction techniques — Principal Component Analysis (PCA), Factor Analysis (FA), and Independent Component Analysis (ICA) — to compress it to two dimensions. The experiment compares how each technique organizes the data and discusses how dimensionality reduction in general mitigates the curse of dimensionality.

#### 2. Experimental Planning (Methodology)

- Load the Wine dataset (178 samples, 13 features such as alcohol, malic acid, flavanoids, colour intensity; 3 cultivar classes).
- Standardize all 13 features using StandardScaler, since PCA/FA/ICA are all scale-sensitive.
- Fit PCA with 2 components and record the explained variance ratio of each component.
- Fit a full PCA (all 13 components) to compute the cumulative explained-variance curve and determine how many components are needed to retain 90% of the information.
- Fit Factor Analysis with 2 latent factors, which models each feature as a linear combination of shared latent factors plus independent noise.
- Fit FastICA with 2 independent components, which seeks statistically independent (rather than merely uncorrelated) latent signals.
- Visualize all three 2-D projections side by side, coloured by the true wine cultivar, to compare how well each technique separates the natural classes.

#### 3. Variable Identification

| Variable                                              | Type                               | Role in Experiment                                                   |
|-------------------------------------------------------|------------------------------------|----------------------------------------------------------------------|
| 13 chemical features (alcohol, ash, flavanoids, etc.) | Independent / Input features       | Original high-dimensional measurements                               |
| Dimensionality-reduction technique (PCA / FA / ICA)   | Independent / Manipulated variable | The factor being compared                                            |
| Number of components (2)                              | Controlled variable                | Fixed at 2 for all three methods for a fair, visualizable comparison |



| Extracted components / factors | Dependent / Output variable     | The new, reduced-dimensional representation of each sample               |
|--------------------------------|---------------------------------|--------------------------------------------------------------------------|
| Wine cultivar class (0-2)      | Reference / Validation variable | Not used for reduction; used only to colour and assess visual separation |

#### 4. Data Collection and Analysis

The Wine dataset (178 samples  $\times$  13 standardized features, 3 cultivar classes) was reduced to two dimensions using each of the three techniques. PCA's first two components explained 36.2% and 19.2% of the variance respectively (55.4% combined). A full-dimensionality PCA showed that 8 of the original 13 components are required to retain 90% of the total variance, which directly demonstrates the curse of dimensionality: the last several raw features individually carry very little unique information, yet a naive model using all 13 raw features would still pay the full statistical and computational cost of that dimensionality.

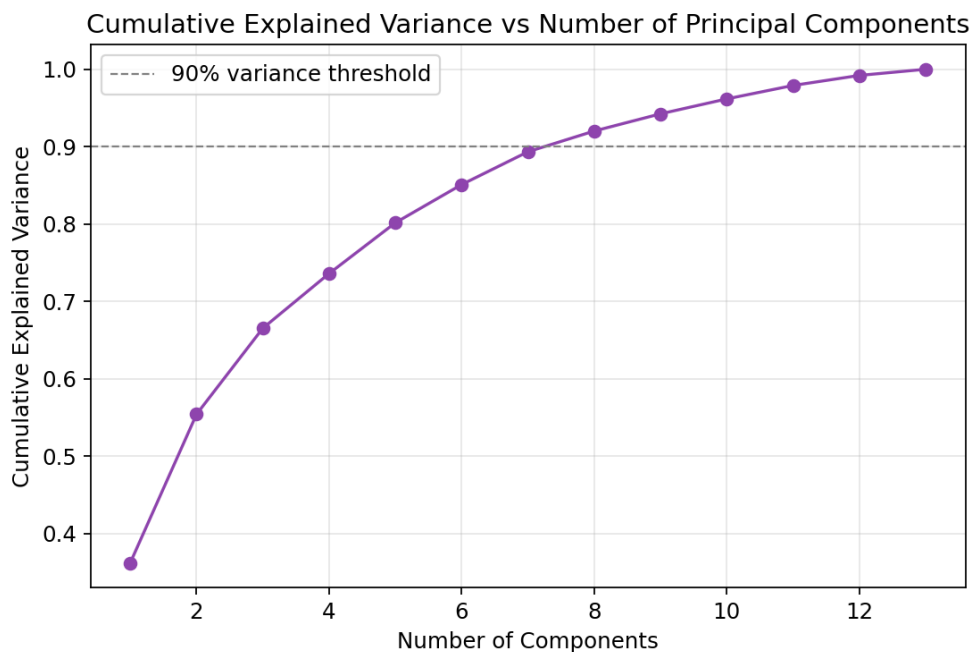
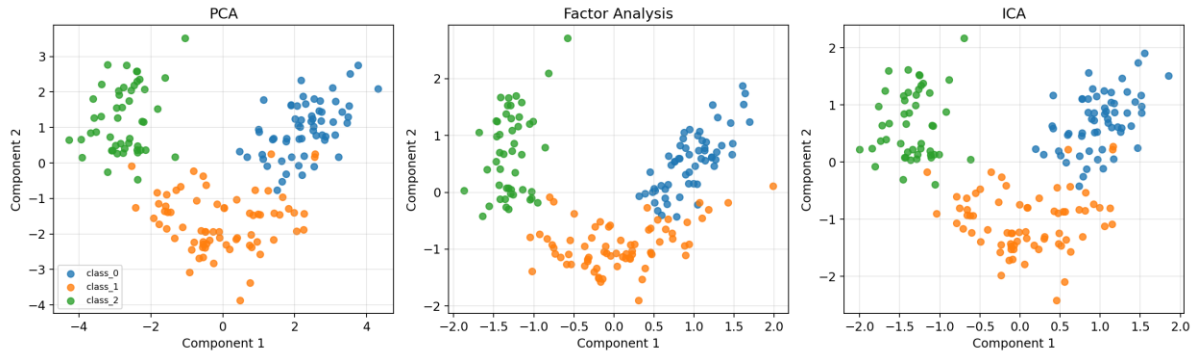


Figure 3.1 — Cumulative explained variance vs number of principal components. Roughly 8 of 13 components are needed to cross the 90% variance threshold, showing that later components add diminishing information.



V

PCA produces orthogonal components ordered by variance and gives the cleanest, most linearly-separated view of the three cultivars, because the wine classes are largely distinguished by directions of maximum variance in the chemical measurements (e.g., alcohol content, flavanoids). Factor Analysis produces a projection that is visually similar to PCA but explicitly models feature-specific noise separately from shared latent structure, so it is less influenced by features that are mostly noise for a given sample. ICA, which optimizes for statistical independence rather than variance or shared covariance, produces axes that are less obviously aligned with class separation in this dataset, since the wine classes are not primarily driven by independent non-Gaussian source signals; ICA is generally better suited to problems like source/signal separation than to maximizing class discriminability.

## 5. Conclusion

All three techniques successfully compressed the 13-dimensional Wine dataset into 2 interpretable dimensions, directly addressing the curse of dimensionality by removing redundant/correlated information: PCA needs only 8 of 13 original dimensions to preserve 90% of the variance, and just 2 components already explain over half of it while visually separating the three cultivars well. PCA and Factor Analysis produced qualitatively similar, well-separated projections because both exploit shared linear covariance structure among correlated chemical features, while ICA produced a less class-discriminative projection because it optimizes for independence rather than variance or explained covariance. In practice, PCA is the most effective and simplest choice here for exploratory visualization and noise/redundancy reduction, FA is preferable when explicit modelling of measurement noise is important, and ICA remains valuable primarily when the true underlying sources are believed to be statistically independent signals.